

PYTHON CONVERSATIONAL LOOP CHATBOT

30-Minute Task Assignment Report

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Task Duration	30 Minutes

TASK OVERVIEW

Build a simple rule-based conversational chatbot in Python that interacts continuously with the user, recognizes different messages, provides appropriate responses, and allows the user to exit gracefully.

REQUIREMENTS CHECKLIST

- ✓ Ask the user's name before the main conversation
- ✓ Use while True: to keep the conversation running
- ✓ Use input() to receive messages from user
- ✓ Clean the input using .lower().strip() methods
- ✓ Recognize at least two greetings (hello, hi)
- ✓ Answer at least three different questions or topics
- ✓ Use if / elif / else for decision making
- ✓ Use break to exit when user types bye or exit
- ✓ Use the user's name in at least one response
- ✓ Provide useful response for unknown messages using else

PYTHON CHATBOT IMPLEMENTATION

Complete Code:

```
# Rule-Based Python Chatbot
# Author: M.Venkata Karthik
# Date: 11-09-2026

def chatbot():
    # Greeting message
    print("=" * 50)
    print("Welcome to PyBot - Your AI Assistant")
    print("=" * 50)

    # Get user's name
    user_name = input("\nWhat is your name? ").strip().title()

    if not user_name:
```

```

user_name = "Friend"

print(f"\nNice to meet you, {user_name}! ■")
print("Type 'bye' or 'exit' to end the conversation.\n")

# Main conversation loop
while True:
    # Get user input and clean it
    user_input = input(f"{user_name}: ").lower().strip()

    # Exit conditions
    if user_input in ['bye', 'exit', 'quit']:
        print(f"PyBot: Goodbye, {user_name}! Have a great day! ■")
        break

    # Greeting responses
    elif 'hello' in user_input or 'hi' in user_input:
        print(f"PyBot: Hello {user_name}! Nice to chat with you! ■")

    # Question: How are you?
    elif 'how are you' in user_input or 'how do you do' in user_input:
        print(f"PyBot: I'm doing great, {user_name}! Thanks for asking! ■")

    # Question: What is your name?
    elif 'your name' in user_input or 'who are you' in user_input:
        print("PyBot: I'm PyBot, a rule-based chatbot built in Python!")

    # Question: What can you do?
    elif 'what can you do' in user_input or 'capabilities' in user_input:
        print("PyBot: I can chat, answer basic questions, and have fun conversations!")

    # Question: Tell me a joke
    elif 'joke' in user_input or 'funny' in user_input:
        print("PyBot: Why do programmers prefer dark mode? Because light attracts bugs! ■")

    # Question: What is AI?
    elif 'artificial intelligence' in user_input or 'what is ai' in user_input:
        print("PyBot: AI is the simulation of intelligent behavior by computer systems!")

    # Unknown input response
    else:
        print(f"PyBot: Interesting! Could you tell me more about that, {user_name}?")

# Run the chatbot
if __name__ == "__main__":
    chatbot()

```

TEST RESULTS & EXECUTION

Test Input	Expected Behavior	Result
hello	Greeting response with name	✓ PASS
HELLO	Case-insensitive greeting	✓ PASS
hi there	Greeting with "hi" keyword	✓ PASS
how are you	Personal question response	✓ PASS
what is ai	AI definition response	✓ PASS
tell me a joke	Joke response with humor	✓ PASS
unknown topic	Graceful fallback response	✓ PASS
bye	Exit and close program	✓ PASS

KEY FEATURES IMPLEMENTED

- 1. Name Collection:** Chatbot asks for user's name at startup and uses it throughout the conversation for personalization.
- 2. Input Cleaning:** All user inputs are converted to lowercase and whitespace is stripped for consistent pattern matching.
- 3. Pattern Recognition:** Uses 'in' operator to detect keywords within longer sentences (e.g., 'hello bot' triggers greeting).
- 4. Multiple Responses:** Handles 6+ different conversation topics with appropriate responses.
- 5. Graceful Exit:** User can exit using 'bye', 'exit', or 'quit' commands without errors.
- 6. Fallback Logic:** Unknown inputs receive a friendly, context-aware response.
- 7. Loop Control:** Uses while True with break statement for continuous conversation.

CHALLENGES & SOLUTIONS

Challenge 1 - Case Sensitivity: User input 'HELLO' vs 'hello' should trigger the same response. **Solution:** Used `.lower()` to normalize all input.

Challenge 2 - Flexible Matching: Phrases like 'hey hello there' should trigger greeting. **Solution:** Used 'in' operator instead of exact string matching.

Challenge 3 - Program Termination: Need clean exit without errors. **Solution:** Used break statement inside if condition checking for exit keywords.

Challenge 4 - Personalization: Make responses feel personal. **Solution:** Store user name and use in responses and prompts.

HOW TO RUN THE CHATBOT

1. Save the code to a file named 'chatbot.py'
2. Open terminal/command prompt in the file directory
3. Run: `python chatbot.py`
4. Enter your name when prompted

5. Start chatting! Type your messages and press Enter
6. Type 'bye' or 'exit' to quit

CONCLUSION

This rule-based chatbot successfully demonstrates core Python concepts including loops, conditionals, string methods, and user interaction. The chatbot is flexible, user-friendly, and handles edge cases gracefully. All 10 requirements have been met and tested thoroughly.

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